

Programming Assignment V0: EVEN–ODD Membership

Problem Description

Write a Java program that decides, for each input string, whether the *entire* string belongs to the language EVEN–ODD. EVEN–ODD consists of all strings over $\{a, b\}$ having an even number of 'a' and an odd number of 'b'. A string is in EVEN–ODD only if every character is 'a' or 'b' and both parity conditions hold.

Program Interface

Command line: `java EvenOddMember <inputFile> <outputFile>`

- The program will be invoked with exactly two arguments: input filename and output filename. You do not need to check the number of arguments; if incorrect, it may crash.
- Input file format:
 - Line 1: a single integer N ($N \geq 0$)
 - Lines 2.. $N+1$: each line contains a single string (may contain characters other than 'a' or 'b').
- Output file format: N lines of output. For each input string, in order, write a single line containing `true` if the whole string is in EVEN–ODD, or `false` otherwise.

Examples

Example input file (input.txt):

```
6
b
aba
bbb
ab
aaa
xxbxx
```

Example output file (output.txt):

```
true
true
true
false
false
false
```

Requirements

- You must use `java.util.regex.Pattern` and `java.util.regex.Matcher`.

- Your program must decide membership by matching against a regular expression, not by counting characters. Tallying the number of 'a's and 'b's to determine membership — rather than applying the pattern — is not allowed, even though it would produce the same answers.
- Membership is a whole-string test: the string is in EVEN–ODD only if the pattern matches the *entire* input, not merely some part of it. Use the `Matcher` method that tests the entire text — recall the difference between `matches()`, `lookingAt()`, and `find()` from the regular-expression slides.
- A string containing any character other than 'a' or 'b' is not in EVEN–ODD and must yield `false`.